VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY



Pham Nhat Minh

RESEARCH ON ENHANCING VISION-LANGUAGE MODEL FOR DOCUMENT KEY INFORMATION EXTRACTION

UNDERGRADUATE GRADUATION THESIS
Major: Information Systems

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Abstract

Key Information Extraction (KIE) on Visually Rich Documents (VRDs)—such as administrative forms, commercial invoices, and retail receipts—aims to map a raw page image and OCR-detected subword tokens with 2D bounding boxes into structured, typed entity spans for downstream enterprise automation. Although multimodal Transformers like LayoutLMv3 jointly encode text, 2D coordinates, and visual patch tokens, standard token-level fine-tuning exhibits four structural failure modes: (1) cross-column coordinate ambiguity from flat (x_0, y_0, x_1, y_1) embeddings, (2) intra-segment label jitter and long-range key-value disconnection caused by isolated subword classification, (3) intra-line boundary bleeding where adjacent key and value spans on the same horizontal line merge into a single entity, and (4) cross-modal representation drift and background noise sensitivity under scanning artifacts or watermarks.

To resolve these bottlenecks without pre-training from scratch, I propose a structure-aware LayoutLMv3 architecture integrating four synergistic components: Hierarchical Position Encoding (HPE), a Segment Context Module (SCM), an Intra-Line Boundary Loss ($\mathcal{L}_{\text{bound}}$), and a calibrated Cross-Modal Contrastive Loss ($\mathcal{L}_{\text{cont}}$). HPE clusters word boxes into reading lines, vertical columns, and local blocks to inject explicit layout topology prior to self-attention. SCM mean-pools token states into line-level segment vectors, models inter-segment reading context via a ReZero-gated Transformer layer, and broadcasts segment consensus back to constituent tokens alongside a segment-onset indicator embedding. During training, $\mathcal{L}_{\text{bound}}$ supervises adjacent intra-line token pairs to enforce sharp entity transitions, while $\mathcal{L}_{\text{cont}}$ aligns pre-SCM token states with detached visual patch anchors via spatial-overlap soft-target InfoNCE, eliminating cross-modal gradient interference and incurring zero inference overhead.

I evaluate my framework across three random seeds ($\{42, 123, 1993\}$) on the

FUNSD and CORD benchmarks using an NVIDIA A100 GPU. On FUNSD, my full 4-module architecture achieves an entity-level F1 score of $92.12 \pm 0.14\%$ on `base` ($+2.20\%$ over the $89.92 \pm 0.85\%$ baseline) and $93.35 \pm 0.18\%$ on `large` ($+2.11\%$ over the $91.24 \pm 0.61\%$ baseline), while reducing span boundary errors by 82.6%. On CORD, my model attains $97.28 \pm 0.12\%$ F1 on `base` and $97.62 \pm 0.08\%$ F1 on `large`. Systematic leave-one-out ablations confirm that every proposed module improves over the baseline, with SCM providing the largest single performance contribution.

Keywords: Document Key Information Extraction, Visually Rich Documents, LayoutLMv3, Hierarchical Position Encoding, Segment Context Module, Intra-Line Boundary Loss, Cross-Modal Contrastive Learning.

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Lastly, I want to thank my parents, family, and friends. Their constant belief in me, patience, and care kept me motivated through many late nights of model training and debugging.

Hanoi, December 2026
Student

Pham Nhat Minh

Declaration

I hereby declare that the research methodology, experimental implementation, and analytical results presented in this graduation thesis are my own work, conducted under the scientific supervision of PhD. Pham Thi Quynh Trang.

All external literature, datasets, and prior research works referenced in this thesis have been properly and clearly cited without any unauthorized copying or plagiarism. Should any violation be found, I take full responsibility in accordance with the regulations of the University of Engineering and Technology – Vietnam National University, Hanoi.

Student

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LIST OF ACRONYMS

Term	**Acronym**	**Description**
Key Information Extraction	KIE	Extracting structured semantic entities and key-value fields from documents
Visually Rich Documents	VRDs	Documents combining text, 2D spatial layout, and visual styling (forms, receipts, invoices)
Document Artificial Intelligence	Document AI	Automated reading, structural parsing, and extraction of document data
Optical Character Recognition	OCR	Detecting text regions and transcribing character sequences from document images
Hierarchical Position Encoding	HPE	Multi-scale spatial encoding across reading lines, vertical columns, and local blocks
Segment Context Module	SCM	Module that pools intra-segment tokens and models inter-segment reading-order context
Intra-Line Boundary Loss	$\mathcal{L}_{\text{bound}}$	Auxiliary binary cross-entropy loss supervising entity transitions between adjacent intra-line tokens
Cross-Modal Contrastive Loss	$\mathcal{L}_{\text{cont}}$	Soft-target InfoNCE objective aligning pre-SCM text states with detached visual patch anchors

Term	Acronym	Description
Inside-Outside-Beginning	IOB2 / BIO	Sequence-labeling tag scheme marking entity onsets, continuations, and background tokens
Masked Language Modeling	MLM	Self-supervised pre-training objective reconstructing masked subword tokens
Masked Image Modeling	MIM	Self-supervised pre-training objective reconstructing discrete tokens of masked image patches
Word-Patch Alignment	WPA	Pre-training objective predicting whether the visual patch corresponding to a word is unmasked
Enterprise Resource Planning	ERP	Enterprise software systems for accounting, procurement, and supply chain management
Know Your Customer	KYC	Regulatory identity and financial document verification in banking and FinTech

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Preface

Automating structured data extraction from scanned administrative forms, tax declarations, commercial invoices, and retail receipts is a critical requirement across modern enterprise workflows in finance, healthcare, logistics, and public administration. Unlike plain-text natural language corpora, Visually Rich Documents (VRDs) convey semantics jointly through lexical content, two-dimensional geometric layout, and visual typography. A numeric string on a receipt derives its meaning—whether unit price, tax amount, or grand total—from its tabular row alignment, column header, and font prominence.

While multimodal Vision-Language Transformers such as LayoutLMv3 unify subword tokens, 2D bounding coordinates, and linear image patch projections within a single backbone, adapting the baseline model directly to downstream Key Information Extraction (KIE) exposes four recurring structural limitations:

1. **Cross-column coordinate ambiguity:** Embedding (x_0, y_0, x_1, y_1) as flat coordinates provides no explicit inductive bias for reading lines, multi-column boundaries, or paragraph blocks.

2. **Intra-segment label jitter and key-value disconnection:** Classifying each BPE subword token in isolation frequently yields conflicting entity tags inside a single multi-token phrase and fails to propagate semantic labels from form keys to standalone values.

3. **Intra-line boundary bleeding:** When a field prompt and its filled value appear consecutively on the same horizontal line (e.g., `TO: George Baroody`), standard token cross-entropy frequently merges both spans into a single entity.

4. **Cross-modal representation drift under visual clutter:** Fine-tuning solely with token classification loss allows visual patch representations to

drift or attract spurious attention from scanning borders, hole punches, and background watermarks.

To address these challenges, I design and evaluate a structure-aware LayoutLMv3 architecture comprising Hierarchical Position Encoding (HPE), a ReZero-gated Segment Context Module (SCM) with segment-onset embeddings, an Intra-Line Boundary Loss, and a calibrated Cross-Modal Contrastive Loss with detached visual patch anchors. Both auxiliary losses regularize representation learning exclusively during training and are discarded at test time, preserving the exact inference latency and memory footprint of the backbone classifier.

This thesis is organized into five chapters:

- **Chapter 1 (Introduction):** Formulates the VRD Key Information Extraction problem, explicitly specifies the model's multimodal inputs and structured entity outputs, outlines real-world industrial applications, and summarizes my core contributions.
- **Chapter 2 (Background and Related Work):** Reviews multimodal document representation, 2D spatial-aware attention, the LayoutLM architecture family, and recent structural and contrastive tuning methods.
- **Chapter 3 (Proposed Methodology):** Presents the mathematical formulation and architectural design of HPE, SCM, Intra-Line Boundary Loss, and calibrated Cross-Modal Contrastive Loss.
- **Chapter 4 (Experiments and Results):** Details my experimental setup on an NVIDIA A100 GPU, benchmark statistics on FUNSD and CORD, contrastive calibration findings, multi-seed evaluation results, leave-one-out ablation studies, and boundary error analysis.
- **Chapter 5 (Conclusion):** Summarizes the main findings of my research, analyzes current architectural limitations, and outlines concrete directions for future work.

Chapter 1

Introduction

1.1 Background and Core Problem

Key Information Extraction (KIE) from Visually Rich Documents (VRDs)—including administrative questionnaires, commercial invoices, tax declarations, and retail receipts—is a core problem in Document Artificial Intelligence [1]. Unlike standard natural language streams where semantics are governed by one-dimensional word order, VRDs encode meaning at the intersection of lexical content, two-dimensional geometric layout, and visual typography [2, 3]. Whether a numeric token represents an item count, a unit price, a tax deduction, or a grand total is determined by its column alignment, its spatial proximity to key headers, and its visual emphasis (e.g., boldface or enlarged font).

Pre-trained multimodal Transformers, most notably LayoutLMv3 [3], unify BPE text embeddings, 2D bounding box coordinates, and linear image patch projections within a single self-attention backbone. Despite strong pre-training representations, directly fine-tuning a token-level linear classifier on top of LayoutLMv3 fails to address four concrete structural bottlenecks encountered on real-world forms and receipts:

1. **Cross-Column Coordinate Ambiguity:** Baseline LayoutLMv3 embeds normalized bounding coordinates (x_0, y_0, x_1, y_1) independently along each axis. Tokens located in separate vertical columns that share similar y-coordinates receive nearly identical vertical embeddings, while tokens in multi-row paragraphs lack explicit block grouping. Consequently, self-attention must infer reading lines, column boundaries, and logical blocks entirely from raw

coordinate numbers.

2. **Intra-Segment Label Jitter and Key-Value Disconnection:** Classifying each subword token independently without an explicit line-segment representation frequently causes prediction drift across multi-token phrases (e.g., switching between `I-ANSWER` and `I-QUESTION` mid-sentence). Furthermore, standalone numeric values on receipts lack intrinsic lexical cues and require inter-segment context propagation from neighboring key segments (such as `Total`, `Cash`, or `Change`).

3. **Intra-Line Entity Span Bleeding:** In compact forms and thermal receipts, distinct semantic entities frequently appear side by side on the same horizontal reading line—for example, an inline prompt followed immediately by its value (`TO: George Baroody`) or a dense receipt row containing quantity, item name, and price. Standard token-wise cross-entropy applies no pairwise transition penalty between adjacent intra-line tokens, causing neighboring spans to merge into a single erroneous entity.

4. **Cross-Modal Representation Drift and Background Clutter:** Although LayoutLMv3 incorporates Word-Patch Alignment during pre-training, downstream fine-tuning relies solely on text-token classification loss. Without continued cross-modal supervision, visual patch representations drift or attract spurious attention from scanning borders, hole-punch margins, and background watermarks [4, 5]. Conversely, naive symmetric contrastive objectives backpropagate noisy visual gradients into the segment classifier, destabilizing token boundaries.

The objective of my thesis is to solve these four structural failure modes within a unified, parameter-efficient architecture that requires no pre-training from scratch and adds zero latency or memory overhead at inference time.

1.2 Problem Formulation: Input, Output, and Objective

I formulate Visually Rich Document Key Information Extraction as a multimodal structure-aware sequence labeling and span extraction task. The exact

inputs, intermediate topological descriptors, and structured outputs of my model are defined below.

1.2.1 Model Input

Given a scanned or photographed document page paired with an upstream Optical Character Recognition (OCR) transcript, each input sample $\mathcal{D}$ supplied to the model consists of two synchronized modalities:

$$\mathcal{D} = \left(\mathbf{I},\ \left\{ \left(w_i,\ \mathbf{b}_i,\ s_i,\ (id_l^{(i)}, id_c^{(i)}, id_b^{(i)}) \right) \right\}_{i=1}^{L} \right), \tag{1.1}$$

where:

- **Visual Page Image (I):** An RGB document image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, which is resized to 224×224 pixels and partitioned into $M = 14 \times 14 = 196$ non-overlapping 16×16 visual patches $\mathbf{P} = (\mathbf{p}_1, \ldots, \mathbf{p}_M)$.
- **Subword Token Sequence (w_i):** A sequence of L Byte-Pair Encoding (BPE) subword tokens $w_i \in \mathcal{V}$ ($L \leq 512$) sorted in reading order.
- **Normalized 2D Bounding Boxes ($\mathbf{b}_i$):** For each token w_i, a 4-tuple of integer coordinates $\mathbf{b}_i = (x_0^{(i)}, y_0^{(i)}, x_1^{(i)}, y_1^{(i)}) \in [0, 1000]^4$ normalized relative to the original page dimensions (W, H), together with derived box width $w^{(i)} = x_1^{(i)} - x_0^{(i)}$ and height $h^{(i)} = y_1^{(i)} - y_0^{(i)}$.
- **OCR Line Segment Index (s_i):** A discrete segment identifier $s_i \in \{0, 1, \ldots, S-1\}$ grouping tokens that belong to the same horizontal OCR text line ($S \ll L$).
- **Hierarchical Layout Indices** $(id_l^{(i)}, id_c^{(i)}, id_b^{(i)})$**:** Deterministic structural indices extracted during preprocessing that assign token w_i to its reading line $id_l^{(i)}$, vertical column slot $id_c^{(i)}$, and local paragraph/field block $id_b^{(i)}$.

1.2.2 Model Output

Given K target semantic entity classes $\mathcal{C} = \{c_1, c_2, \ldots, c_K\}$ (e.g., `HEADER`, `QUESTION`, `ANSWER` on FUNSD, or 30 fine-grained receipt fields such as `menu.nm` and `total.total_price` on CORD), the token-level label space $\mathcal{Y}$ follows the IOB2 scheme of cardinality $|\mathcal{Y}| = 2K + 1$:

$$\mathcal{Y} = \{\texttt{O}\} \cup \bigcup_{k=1}^{K} \{\texttt{B-}c_k,\ \texttt{I-}c_k\}\,. \tag{1.2}$$

At inference time, the model $f_\Theta(\mathcal{D})$ produces two hierarchical levels of output:

1. **Token-Level Logits and Tag Sequence:** For each valid word-initial subword token $i \in \{1, \ldots, L\}$, the classification head outputs a logit vector $\mathbf{z}_i \in \mathbb{R}^{|\mathcal{Y}|}$ and predicted tag $\hat{y}_i = \arg\max_{y \in \mathcal{Y}} z_{i,y}$, forming the predicted tag sequence $\hat{\mathbf{Y}} = (\hat{y}_1, \ldots, \hat{y}_L) \in \mathcal{Y}^L$.

2. **Structured Semantic Entity Spans:** Contiguous sequences of `B-`c_k followed by zero or more `I-`c_k tokens are deterministically decoded into a set of structured entity tuples:

$$\hat{\mathcal{E}} = \left\{ \left(i_{\text{start}},\, i_{\text{end}},\, c_k,\, \text{text}(i_{\text{start}} : i_{\text{end}}),\, \mathbf{b}_{\text{union}}(i_{\text{start}} : i_{\text{end}}) \right) \right\}, \tag{1.3}$$

   where each extracted tuple contains the token span boundaries, semantic category $c_k \in \mathcal{C}$, reconstructed surface string, and merged bounding box—directly serializable into JSON records for downstream database or ERP ingestion.

1.3 Real-World Applications

Because my proposed architecture enhances structural accuracy and boundary precision while preserving the exact inference footprint of a single forward pass, it can be deployed directly across five high-impact industrial domains:

1. **Enterprise ERP, Accounting, and Accounts Payable Automation:** Organizations process millions of multi-vendor invoices, purchase orders, and retail receipts whose layouts vary arbitrarily across suppliers. My model extracts line-item tables (`menu.nm`, `menu.cnt`, `menu.unitprice`, `menu.price`), tax breakdowns (`sub_total.tax_price`), and settlement totals (`total.total_price`, `total.cashprice`) directly into Enterprise Resource Planning (ERP) systems such as SAP, Oracle NetSuite, or Odoo, eliminating manual data entry and enabling automated three-way matching between purchase orders, goods receipts, and invoices.

2. **Banking, FinTech KYC, and Automated Underwriting:** Retail and commercial banks evaluate customer onboarding applications, tax returns, payroll stubs, and mortgage questionnaires characterized by dense two-column grids and inline key-value fields. By resolving cross-column coordinate

ambiguity via HPE and preventing inline key-value merging via $\mathcal{L}_{\text{bound}}$, the model reliably separates field prompts (`QUESTION`) from applicant entries (`ANSWER`), accelerating Know Your Customer (KYC) verification and automated credit risk scoring.

3. **Healthcare Clinical Records and Insurance Claims Adjudication:** Hospitals and health insurers handle scanned patient intake questionnaires, diagnostic laboratory reports, and medical reimbursement claims that frequently suffer from scan skew, stamp overlays, and hole-punch artifacts. The calibrated Cross-Modal Contrastive Loss ($\mathcal{L}_{\text{cont}}$) anchors text representations to true local typography while repelling margin and watermark noise, enabling high-precision extraction of patient identifiers, diagnostic codes, and itemized treatment charges.

4. **Global Supply Chain, Logistics, and Customs Clearance:** International freight forwarders and customs brokerages parse bills of lading, commercial packing lists, and certificates of origin containing nested multi-line shipper/consignee blocks alongside dense cargo tables. The Segment Context Module (SCM) maintains semantic consistency across multi-line address blocks while accurately binding column headers to tabular cargo weights and Harmonized System (HS) tariff codes.

5. **E-Government and Legal Archival Digitization:** Public sector agencies digitize historical civil registries, land deeds, and regulatory filings into searchable digital archives. Because the auxiliary boundary and contrastive heads are active only during training, the deployed checkpoint executes at full throughput on standard enterprise GPUs or quantized ONNX/TensorRT runtimes for large-scale batch digitization.

1.4 Summary of Contributions

The main contributions of my graduation thesis are summarized as follows:

1. **Hierarchical Position Encoding (HPE):** I design a deterministic spatial clustering pipeline that decomposes 2D word boxes into reading lines (id_l), vertical columns (id_c), and logical blocks (id_b), projecting their concatenated embeddings into the token stream prior to multimodal self-attention.

2. **Segment Context Module (SCM) with ReZero Gating and Onset Embeddings:** I introduce a lightweight segment-level Transformer encoder over mean-pooled OCR line representations, stabilized by a learnable ReZero residual gate g and paired with a binary segment-onset indicator embedding $\mathbf{e}_{\text{first}}(f_i)$ to eliminate intra-segment label jitter and propagate key-value context across lines.

3. **Intra-Line Boundary Loss ($\mathcal{L}_{\text{bound}}$):** I formulate an auxiliary pairwise binary classification objective over consecutive tokens within the same reading line ($s_i = s_{i+1}$), sharpening entity transitions and cutting span boundary errors on FUNSD by 82.6%.

4. **Calibrated Cross-Modal Contrastive Loss ($\mathcal{L}_{\text{cont}}$):** I diagnose why naive symmetric contrastive tuning degrades token boundaries on smaller backbones and resolve this conflict by aligning pre-SCM token states against detached visual patch anchors (stop-grad) using a spatial-overlap soft-target InfoNCE objective ($\tau = 0.10$), turning cross-modal alignment into a consistent performance gain across all model scales.

5. **Rigorous Multi-Seed Evaluation and Ablation Study:** I evaluate 24 model configurations across three random seeds ($\{42, 123, 1993\}$) on an NVIDIA A100 GPU on FUNSD and CORD. My full 4-module architecture achieves $92.12 \pm 0.14\%$ F1 on FUNSD `base` (+2.20% over baseline) and $93.35 \pm 0.18\%$ F1 on FUNSD `large` (+2.11% over baseline), while reaching $97.28 \pm 0.12\%$ and $97.62 \pm 0.08\%$ F1 on CORD `base` and `large`, respectively.

1.5 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2:** Reviews multimodal document representation, 2D spatial-aware attention, the LayoutLM architecture family, and related structural and contrastive tuning methods.

- **Chapter 3:** Details the mathematical formulation and architectural design of HPE, SCM, Intra-Line Boundary Loss, and calibrated Cross-Modal Contrastive Loss.

- **Chapter 4:** Describes the experimental setup, dataset properties, contrastive calibration analysis, multi-seed benchmark results, 24-configuration leave-one-out ablation studies, and qualitative boundary error analysis.
- **Chapter 5:** Summarizes the thesis findings, analyzes current limitations, and proposes future research directions.

Chapter 2

Background and Related Work

2.1 Multimodal Document Representation

2.1.1 OCR Serialization and Subword Alignment in VRDs

Unlike plain text, Visually Rich Documents (VRDs) require joint modeling of lexical tokens, 2D spatial geometry, and visual rendering [1]. Standard pipelines first apply Optical Character Recognition (OCR) to detect word-level bounding boxes $\tilde{\mathbf{b}}_i$ and transcribe character strings, which are subsequently normalized to the integer coordinate domain $[0, 1000]^4$ (Equation 1.1) to ensure scale invariance across heterogeneous scan resolutions.

When Byte-Pair Encoding (BPE) [6, 7] splits a single OCR word into $m > 1$ subword tokens $(w_{i,1}, \ldots, w_{i,m})$, all constituent subwords inherit the parent word's normalized bounding box $\mathbf{b}_i$ and OCR line segment index s_i. Under standard IOB2 sequence labeling [8], cross-entropy supervision and `seqeval` entity decoding are restricted strictly to the initial subword $w_{i,1}$, while continuation subwords $(w_{i,2}, \ldots, w_{i,m})$ are masked with label index -100 so that subword fertility does not skew gradient magnitudes or boundary evaluation.

2.1.2 2D Spatial-Aware Multimodal Attention

To fuse textual and visual streams within a single Transformer [9], modern document backbones concatenate L text token embeddings with $M + 1$ visual patch embeddings into a unified sequence $\mathbf{H} \in \mathbb{R}^{(L+M+1)\times d}$. Because standard

1D sinusoidal or learned position embeddings capture only serialized reading order, LayoutLMv2 [10] and LayoutLMv3 [3] inject explicit 2D spatial biases into the pre-softmax attention logits of each head h:

$$\mathbf{A}_h = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}} + \mathbf{B}_{\text{rel}}^{\text{1D}} + \mathbf{B}_{\text{rel}}^{\text{2D-}x} + \mathbf{B}_{\text{rel}}^{\text{2D-}y} \right), \tag{2.1}$$

where $\mathbf{B}_{\text{rel}}^{\text{1D}}(i,j)$ encodes relative 1D sequence distance $|i-j|$, while $\mathbf{B}_{\text{rel}}^{\text{2D-}x}(i,j)$ and $\mathbf{B}_{\text{rel}}^{\text{2D-}y}(i,j)$ are learnable bucketed biases indexed by the horizontal and vertical coordinate offsets $(x_0^{(i)} - x_0^{(j)})$ and $(y_0^{(i)} - y_0^{(j)})$. While these pairwise biases capture continuous coordinate deltas, they remain agnostic to discrete structural boundaries—such as whether two horizontally separated tokens belong to the same reading line or to adjacent columns in a multi-column form.

2.2 The LayoutLM Architecture Family

2.2.1 Evolution from LayoutLM to LayoutLMv2

LayoutLM [2] extended BERT [11] by summing 2D coordinate embeddings (x_0, y_0, x_1, y_1) with subword embeddings during Masked Visual-Language Modeling on IIT-CDIP scans, relying on an external Faster R-CNN detector only during downstream fine-tuning. LayoutLMv2 [10] integrated a ResNeXt-FPN convolutional visual backbone directly into pre-training alongside spatial-aware self-attention (Equation 2.1) and cross-modal Text-Image Alignment objectives. However, the convolutional feature extractor introduced substantial parameter overhead and region-alignment bottlenecks.

2.2.2 LayoutLMv3 Backbone

LayoutLMv3 [3] eliminates heavy CNN backbones by adopting Vision Transformer (ViT) linear patch projections [12], as illustrated in Figure 2.1.

Its input representation comprises two streams:

1. **Text Embedding Stream:** Each BPE subword w_i combines its token embedding, 1D position embedding, token-type embedding, and 6-tuple 2D box embedding $(x_0, y_0, x_1, y_1, w, h)$:

$$\mathbf{e}_i^{\text{text}} = \text{LN}\big(\mathbf{e}^{\text{word}}(w_i) + \mathbf{e}^{\text{pos1D}}(i) + \mathbf{e}^{\text{type}}(0) + \mathbf{e}^{\text{pos2D}}(\mathbf{b}_i)\big). \tag{2.2}$$

   Instead of noisy word-level boxes, LayoutLMv3 assigns identical segment-level bounding coordinates to all words within the same OCR text line.

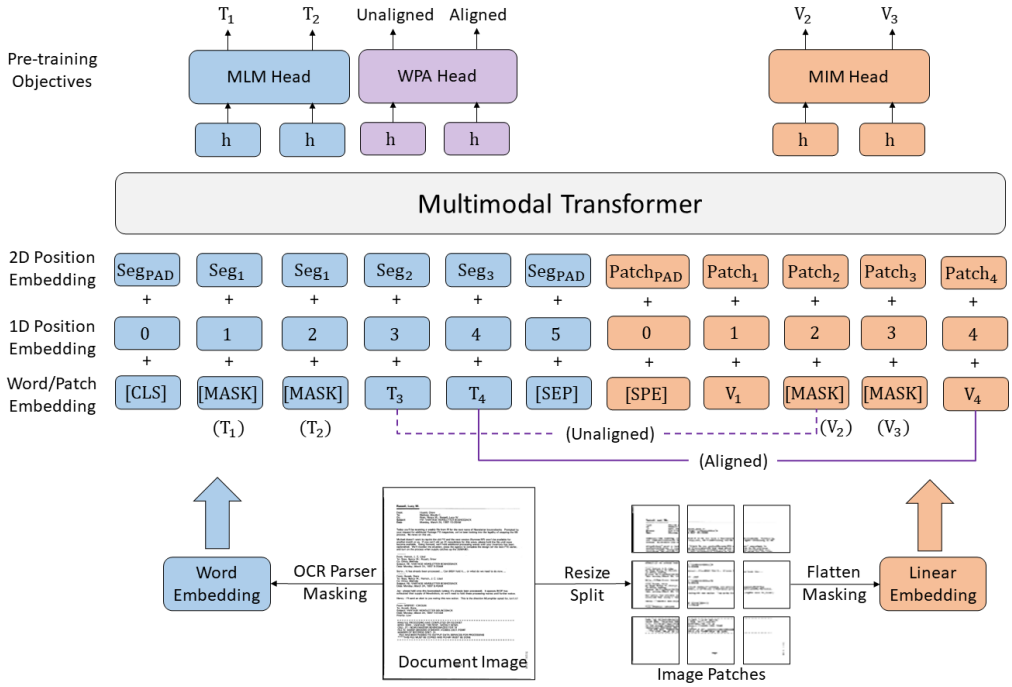


Figure 2.1: Architecture of the LayoutLMv3 backbone, unifying segment-shared 2D text embeddings and linear image patch projections under joint MLM, MIM, and WPA pre-training [3].

2. **Visual Patch Stream:** The 224×224 page image $\mathbf{I}$ is split into $M = 196$ patches of 16×16 pixels, linearly projected to $\mathbb{R}^d$, summed with learnable 1D visual position embeddings, and prepended with $[\texttt{CLS}_{\text{vis}}]$.

Pre-trained on 11 million IIT-CDIP pages using Masked Language Modeling (MLM), Masked Image Modeling (MIM, reconstructing discrete dVAE visual tokens), and Word-Patch Alignment (WPA), LayoutLMv3 provides a strong foundation for document understanding.

2.3 Related Work in Structural and Cross-Modal Document KIE

2.3.1 Topological Layout and Segment Modeling

Recent studies demonstrate that flat 2D coordinate embeddings are insufficient for complex multi-column and tabular layouts. StructuralLM [13] grouped words into shared 2D cell embeddings to capture form regions, while BROS [14] encoded relative spatial offsets between bounding box vertices to improve reading-order

robustness. Graph-based architectures such as FormNet [15] and GraphLayoutLM [16] constructed explicit layout graphs over neighboring text segments to mitigate serialization distortion, and ROAP [5] injected reading-order-aware relative position biases derived from an Adaptive-XY-Gap tree.

Unlike approaches that require custom graph attention kernels or full re-pretraining on millions of pages, my Hierarchical Position Encoding (HPE) and Segment Context Module (SCM) inject multi-scale layout topology (line, column, block) and inter-segment reading context directly into an off-the-shelf LayoutLMv3 checkpoint via ReZero residual gating [17].

2.3.2 Cross-Modal Alignment and Boundary Supervision

During downstream KIE fine-tuning, optimizing token cross-entropy alone leaves the visual patch stream without direct supervision, allowing cross-modal alignment to degrade [4]. AETNet [4] addressed this by introducing patch-token contrastive objectives during fine-tuning, while DocExtractNet [18] combined cross-modal feature enhancement with side-tuning to handle noisy scans. However, my empirical analysis reveals that applying symmetric InfoNCE directly to post-segment token representations creates gradient interference with token boundary classification on smaller backbones. I resolve this limitation by decoupling pre-SCM contrastive alignment with stop-gradient visual anchors from post-SCM classification and pairing it with explicit Intra-Line Boundary supervision.

Chapter 3

Proposed Methodology

3.1 Overview of the Proposed Architecture

To resolve the four structural failure modes of baseline LayoutLMv3 identified in Chapter 1—cross-column coordinate ambiguity, intra-segment label jitter, intra-line span bleeding, and cross-modal representation drift—I design a structure-aware architecture comprising four complementary components:

1. **Hierarchical Position Encoding (HPE):** Clusters token bounding boxes into reading lines (id_l), vertical columns (id_c), and logical blocks (id_b), injecting explicit multi-scale layout topology into the input token embeddings prior to multimodal self-attention.

2. **Segment Context Module (SCM):** Mean-pools token hidden states within each OCR line segment, models inter-segment reading context via a ReZero-gated Transformer layer, and broadcasts segment consensus back to constituent tokens alongside a binary segment-onset embedding.

3. **Intra-Line Boundary Loss ($\mathcal{L}_{\text{bound}}$):** Supervises consecutive intra-line token pairs $(i, i+1)$ with an auxiliary binary transition classifier to enforce sharp cuts between adjacent inline entities.

4. **Calibrated Cross-Modal Contrastive Loss ($\mathcal{L}_{\text{cont}}$):** Aligns pre-SCM token states with detached visual patch anchors via a spatial-overlap soft-target InfoNCE objective, grounding text representations in local visual typography while avoiding gradient interference with post-SCM boundary classification.

Figure 3.1 illustrates the complete end-to-end architecture. Both $\mathcal{L}_{\text{bound}}$ and $\mathcal{L}_{\text{cont}}$ operate strictly as training-time regularizers; during inference, only the linear token classification head is executed, introducing zero additional latency over the segment-enriched backbone.

3.2 Hierarchical Position Encoding (HPE)

Baseline LayoutLMv3 embeds normalized coordinates $(x_0, y_0, x_1, y_1, w, h)$ along flat horizontal and vertical axes, providing no explicit signal indicating whether two tokens at similar y-coordinates reside in the same continuous line or in separate vertical columns. To inject explicit topological structure, I derive three discrete hierarchical indices $(id_l^{(i)}, id_c^{(i)}, id_b^{(i)})$ for each token i from the normalized bounding boxes $\mathbf{b}_i \in [0, 1000]^4$:

1. **Line Index (id_l):** Horizontally adjacent bounding boxes are merged into the same reading line when their vertical intersection exceeds 50% of the shorter box's height:
$$\frac{\min(y_1^{(i)}, y_1^{(j)}) - \max(y_0^{(i)}, y_0^{(j)})}{\min(h^{(i)}, h^{(j)})} \geq 0.50. \tag{3.1}$$
Tokens sharing id_l are ordered monotonically by x_0.

2. **Column Index (id_c):** Within a reading line, a horizontal gap between consecutive tokens exceeding 12% of the normalized page width ($\Delta x > 120$) increments the column slot id_c. This cleanly separates multi-column form fields and right-aligned receipt prices from left-aligned descriptions.

3. **Block Index (id_b):** Vertically consecutive lines within the same column region are grouped into a unified logical block id_b whenever their vertical gap is less than $1.8\times$ the median line height of the page, binding multi-line addresses, paragraphs, and tabular sections.

Each discrete index is mapped to a 128-dimensional learnable embedding via lookup tables $\mathbf{E}_{\text{line}}$, $\mathbf{E}_{\text{col}}$, and $\mathbf{E}_{\text{block}}$. The concatenated 384-dimensional topological vector is projected to the backbone hidden dimension $d \in \{768, 1024\}$ via $\mathbf{W}_{\text{hier}} \in \mathbb{R}^{d\times 384}$ and fused into the input token representation with scale factor $\beta = 1.0$:

$$\mathbf{e}_{\text{hier}}(i) = \mathbf{W}_{\text{hier}} \left[\mathbf{E}_{\text{line}}(id_l^{(i)}) \parallel \mathbf{E}_{\text{col}}(id_c^{(i)}) \parallel \mathbf{E}_{\text{block}}(id_b^{(i)}) \right], \qquad \mathbf{e}_i^{\text{total}} = \mathbf{e}_i^{\text{text}} + \beta\, \mathbf{e}_{\text{hier}}(i). \tag{3.2}$$

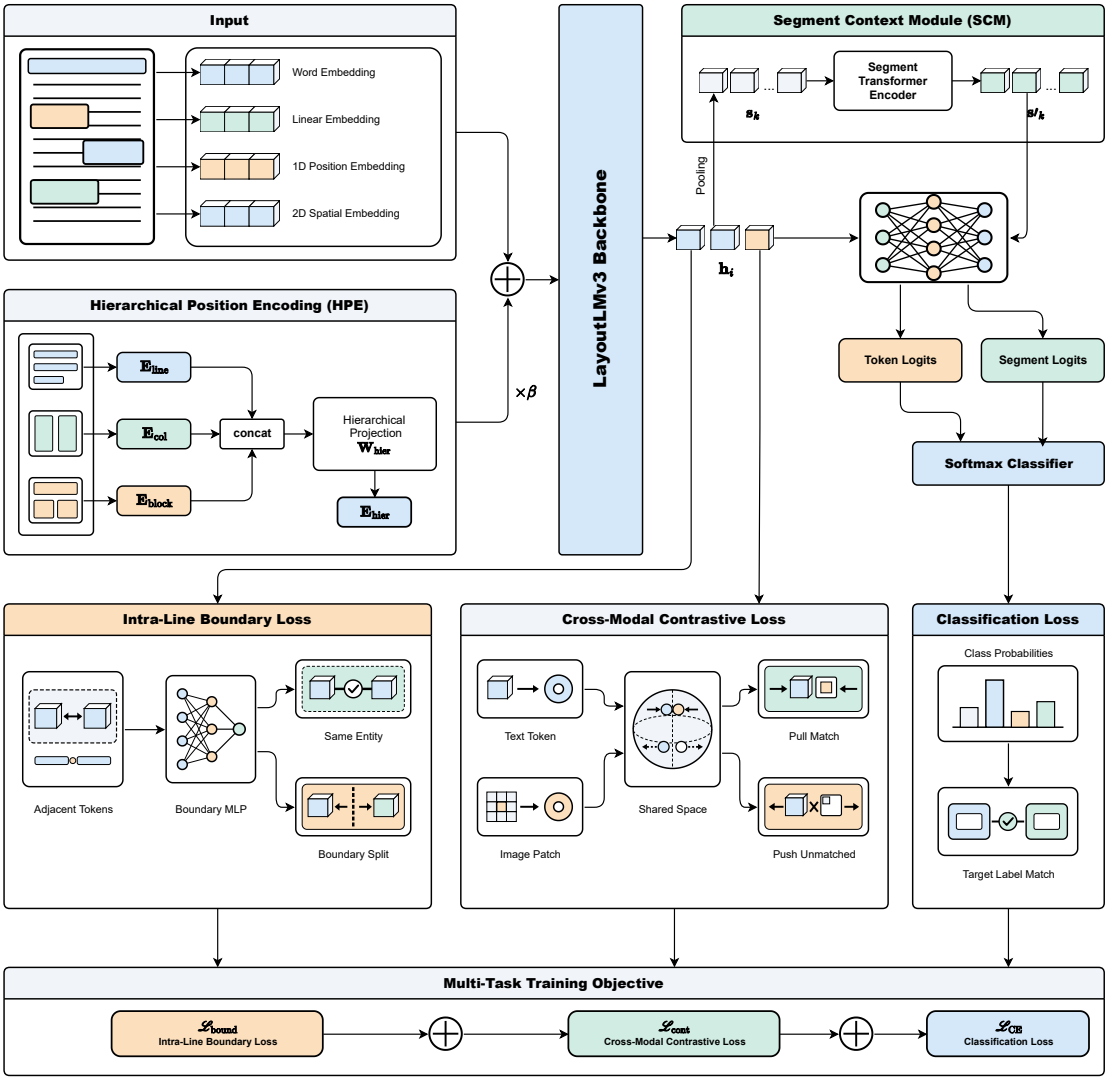


Figure 3.1: End-to-end architecture of my proposed structure-aware LayoutLMv3 framework for Key Information Extraction on Visually Rich Documents, integrating (1) Hierarchical Position Encoding (HPE), (2) Multimodal LayoutLMv3 Backbone, (3) Segment Context Module (SCM) with ReZero gating and onset embeddings, (4) Intra-Line Boundary Loss ($\mathcal{L}_{\text{bound}}$), and (5) Calibrated Cross-Modal Contrastive Loss ($\mathcal{L}_{\text{cont}}$) with detached visual patch anchors.

3.3 Segment Context Module (SCM)

Let $\mathbf{H} = (\mathbf{h}_1, \ldots, \mathbf{h}_L) \in \mathbb{R}^{L \times d}$ and $\mathbf{H}^{\text{img}} = (\mathbf{h}_0^{\text{img}}, \mathbf{h}_1^{\text{img}}, \ldots, \mathbf{h}_M^{\text{img}}) \in \mathbb{R}^{(M+1) \times d}$ denote the contextualized text and visual patch hidden states emitted by the final layer of the LayoutLMv3 encoder. To eliminate intra-segment label jitter within multi-word entities and enable long-range key-value binding across reading lines, SCM processes $\mathbf{H}$ in three stages:

1. **Intra-Segment Mean Pooling:** Given the OCR segment assignment $s_i \in \{0, \ldots, S-1\}$, let $\mathcal{S}_k = \{i \in \{1, \ldots, L\} \mid s_i = k\}$ denote the token indices belonging to segment k. I aggregate constituent token states into a compact segment representation $\mathbf{s}_k \in \mathbb{R}^d$:

$$\mathbf{s}_k = \frac{1}{|\mathcal{S}_k|} \sum_{i \in \mathcal{S}_k} \mathbf{h}_i, \qquad k \in \{0, \ldots, S-1\}. \tag{3.3}$$

2. **ReZero-Gated Inter-Segment Transformer:** To model semantic dependencies across reading lines (e.g., transferring context from a `Total` or `Question` segment to a standalone numeric or answer segment), I add a learnable segment-order embedding $\mathbf{E}_{\text{seg}}(k) \in \mathbb{R}^d$ and pass the sequence $(\mathbf{s}_0, \ldots, \mathbf{s}_{S-1})$ through a lightweight 1-layer bidirectional Transformer encoder (`SegmentTransformerLayer`). To prevent randomly initialized segment attention weights from perturbing pre-trained backbone representations at the start of fine-tuning, I apply a learnable ReZero residual gate $g \in \mathbb{R}$ initialized at $g = 0$ [17]:

$$\mathbf{c}_k = \text{TransformerEncoder}\big(\mathbf{s}_k + \mathbf{E}_{\text{seg}}(k)\big), \qquad \mathbf{s}'_k = \mathbf{s}_k + g \cdot (\mathbf{c}_k - \mathbf{s}_k). \tag{3.4}$$

3. **Segment Broadcast and Onset Indicator Embedding:** When $\mathbf{s}'_{s_i}$ is broadcast back to all tokens $i \in \mathcal{S}_{s_i}$, tokens within the same segment receive an identical contextual shift. To preserve sharp differentiation between entity onset (`B-`) and continuation (`I-`) tokens at segment starts, I compute a binary segment-onset indicator $f_i = \mathbb{I}[i = 1 \ \vee \ s_i \neq s_{i-1}] \in \{0, 1\}$ and look up a learnable onset embedding $\mathbf{e}_{\text{first}}(f_i) \in \mathbb{R}^d$:

$$\mathbf{h}_i^{\text{final}} = \mathbf{h}_i + \mathbf{s}'_{s_i} + \mathbf{e}_{\text{first}}(f_i). \tag{3.5}$$

3.4 Token Classification and Intra-Line Boundary Loss

The segment-enriched token states $\mathbf{h}_i^{\text{final}}$ are projected via a linear classification head $\mathbf{z}_i = \mathbf{W}_{\text{cls}}\mathbf{h}_i^{\text{final}} + \mathbf{b}_{\text{cls}} \in \mathbb{R}^{|\mathcal{Y}|}$ and supervised on valid word-initial tokens $\mathcal{V}_{\text{valid}}$ ($y_i \neq -100$) using standard multi-class cross-entropy loss $\mathcal{L}_{\text{CE}}(\mathbf{Z}, \mathbf{Y})$.

Because $\mathcal{L}_{\text{CE}}$ evaluates each token position independently and SCM broadcasts a shared vector $\mathbf{s}'_k$ to all tokens on line k, adjacent distinct entities sharing the same horizontal line (such as an inline key-value pair `TO: George Baroody` on FUNSD or a dense `cnt–nm–price` item row on CORD) remain vulnerable to boundary bleeding. To enforce sharp intra-line entity cuts, I define the set of valid consecutive intra-line token pairs:

$$\mathcal{P}_{\text{intra}} = \{(i, i+1) \mid s_i = s_{i+1} \ \wedge \ i, i+1 \in \mathcal{V}_{\text{valid}}\}\,. \tag{3.6}$$

For each pair $(i, i+1) \in \mathcal{P}_{\text{intra}}$, a two-layer bottleneck MLP (`boundary_detector`: $2d \to d/2 \to 1$ with GELU activation) predicts the probability of an entity boundary transition:

$$p_{i,i+1}^{\text{bound}} = \sigma\big(\mathbf{W}_2\,\text{GELU}\big(\mathbf{W}_1[\mathbf{h}_i^{\text{final}} \parallel \mathbf{h}_{i+1}^{\text{final}}] + \mathbf{b}_1\big) + b_2\big)\,. \tag{3.7}$$

The ground-truth transition target $t_{i,i+1}^{\text{bound}} \in \{0, 1\}$ is set to 1 if and only if token $i+1$ starts a new entity (`B-` tag) or switches semantic category relative to token i ($y_i \neq y_{i+1}$ and y_{i+1} is not the matching `I-` continuation of y_i). The Intra-Line Boundary Loss $\mathcal{L}_{\text{bound}}$ is computed as the binary cross-entropy between $p_{i,i+1}^{\text{bound}}$ and $t_{i,i+1}^{\text{bound}}$ averaged over $\mathcal{P}_{\text{intra}}$.

3.5 Calibrated Cross-Modal Contrastive Loss

While visual patch tokens capture font weight, text scale, and layout borders, fine-tuning exclusively with textual losses allows cross-modal alignment to degrade or succumb to scanning noise (e.g., background watermarks on CORD receipts or hole-punch margins on FUNSD forms). However, in my initial investigation, applying a naive symmetric InfoNCE loss ($\tau = 0.07, \lambda_{\text{cont}} = 0.10$) directly to the post-SCM representations $\mathbf{h}_i^{\text{final}}$ and single center patches slightly degraded entity F1 on `base` models (-0.33% on FUNSD `base` and -0.36% on CORD `base` relative to the 3-module model).

Diagnosing the gradient dynamics revealed three causes of this conflict: (1) pulling post-SCM token states $\mathbf{h}_i^{\text{final}}$ toward spatially coarse 16×16 visual patches directly opposes the fine-grained token boundary separation enforced by $\mathcal{L}_{\text{bound}}$; (2) backpropagating noisy textual gradients into the visual patch stream without stop-gradient destabilizes pre-trained ViT patch features on small batch sizes; and (3) a multi-character word box often spans multiple visual patches rather than a single center patch.

To resolve these issues, my calibrated Cross-Modal Contrastive Loss ($\mathcal{L}_{\text{cont}}$) introduces three architectural refinements:

1. **Pre-SCM Text Projection:** Contrastive alignment is applied to the pre-SCM backbone token state $\mathbf{h}_i$ rather than $\mathbf{h}_i^{\text{final}}$, regularizing the multimodal backbone while leaving the post-SCM boundary representations unperturbed.

2. **Detached Visual Patch Anchors:** Visual patch representations $\mathbf{v}_p = \text{stop-grad}(\mathbf{h}_p^{\text{img}})$ $(p \in \{1, \ldots, M\})$ are detached from contrastive backpropagation, serving as stable visual anchors before ℓ_2-normalized projection into $\mathbb{R}^{128}$:

$$\mathbf{z}_i^{\text{text}} = \frac{\mathbf{W}_{\text{text}}\mathbf{h}_i}{\|\mathbf{W}_{\text{text}}\mathbf{h}_i\|_2}, \qquad \mathbf{z}_p^{\text{img}} = \frac{\mathbf{W}_{\text{img}}\,\text{stop-grad}(\mathbf{h}_p^{\text{img}})}{\|\mathbf{W}_{\text{img}}\,\text{stop-grad}(\mathbf{h}_p^{\text{img}})\|_2}. \tag{3.8}$$

3. **Spatial-Overlap Soft-Target InfoNCE:** Mapping normalized token box coordinates $\mathbf{b}_i \in [0, 1000]^4$ onto the 14×14 visual patch grid defines the set of overlapping patches $\mathcal{P}^+(i)$ along with the primary center patch $p_c(i)$. I assign unnormalized positive affinity weights $\tilde{w}_{i,p} = \mathbb{I}[p \in \mathcal{P}^+(i)] + \mathbb{I}[p = p_c(i)]$ (giving double weight to the center patch), normalized to a target distribution $q_{i,p} = \tilde{w}_{i,p} / \sum_{p'=1}^{M} \tilde{w}_{i,p'}$. The calibrated contrastive objective optimizes soft-target cross-entropy over patch similarities at temperature $\tau = 0.10$:

$$\mathcal{L}_{\text{cont}} = -\frac{1}{|\mathcal{V}_{\text{valid}}|} \sum_{i \in \mathcal{V}_{\text{valid}}} \sum_{p \in \mathcal{P}^+(i)} q_{i,p} \log \frac{\exp\big((\mathbf{z}_i^{\text{text}})^\top \mathbf{z}_p^{\text{img}} / \tau\big)}{\sum_{p'=1}^{M} \exp\big((\mathbf{z}_i^{\text{text}})^\top \mathbf{z}_{p'}^{\text{img}} / \tau\big)}. \tag{3.9}$$

3.6 Multi-Task Optimization and Zero-Overhead Inference

During fine-tuning, all active modules are optimized jointly in a single end-to-end backward pass:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{bound}}\mathcal{L}_{\text{bound}} + \lambda_{\text{cont}}\mathcal{L}_{\text{cont}}, \tag{3.10}$$

with calibrated loss weights $\lambda_{\text{bound}} = 0.10$ and $\lambda_{\text{cont}} = 0.03$ on FUNSD ($\lambda_{\text{cont}} = 0.01$ on CORD). At inference time, `boundary_detector`, $\mathbf{W}_{\text{text}}$, and $\mathbf{W}_{\text{img}}$ are bypassed entirely, and entity predictions are decoded directly via $\hat{y}_i = \arg\max_y z_{i,y}$.

Chapter 4

Experiments and Results

4.1 Experimental Setup and Evaluation Protocol

I conducted all training and ablation experiments on a cloud server equipped with an NVIDIA A100-SXM4-80GB GPU and 167 GiB of system RAM (Table 4.1). Following standard VRD benchmarks [3], I evaluate performance using strict entity-level Precision (P), Recall (R), and micro-F1 computed via `seqeval` under the IOB2 scheme, where a predicted span is counted as a True Positive only if both its exact token boundaries ($i_{\text{start}}, i_{\text{end}}$) and semantic category c_k match the ground truth. To account for optimization variance, every model configuration is trained and evaluated independently across three random seeds $\mathcal{S} = \{42, 123, 1993\}$, and all tables report Mean $\pm$ Std.

Table 4.1: Hardware environment and software stack used across all experiments.

Component	**Specification / Version**
GPU Accelerator	NVIDIA A100-SXM4-80GB (81,920 MiB HBM2e VRAM)
Host Memory & OS	167 GiB System RAM; Ubuntu Linux 20.04 LTS (x86_64)
Software Stack	Python 3.7.16, PyTorch 1.10.0 (CUDA 11.1, AMP `fp16`), Transformers 4.12.5
Evaluation Protocol	`seqeval` 1.2.2 strict entity-level IOB2 F1 across seeds $\mathcal{S} = \{42, 123, 1993\}$

4.2 Benchmark Datasets and Hyperparameters

I evaluate my framework on two canonical Document AI benchmarks representing distinct layout regimes (Table 4.2):

- **FUNSD** [19]: Comprises 199 noisy scanned administrative and questionnaire forms (149 train, 50 test) with 9,707 entities and 31,485 words extracted from RVL-CDIP. Active entities are categorized into `HEADER` (11.8%), `QUESTION` (44.2%), and `ANSWER` (33.1%), alongside background `OTHER` (10.9%), yielding $|\mathcal{Y}| = 7$ IOB2 tags. FUNSD tests multi-column layout disambiguation, multi-line answer consistency, and inline key-value separation.
- **CORD** [20]: Contains 1,000 Indonesian retail and restaurant receipts (800 train, 100 validation, 100 test) captured under mobile camera distortion, thermal fading, and watermark clutter. It defines 30 fine-grained classes across 5 super-categories (`menu.*`, `sub_total.*`, `total.*`, `void_menu.*`, `etc.*`), yielding $|\mathcal{Y}| = 61$ IOB2 tags.

Table 4.2: Summary statistics of the FUNSD and CORD benchmarks.

Dataset	**Domain**	**Train**	**Val**	**Test**	**Tags**	**Target Entity Categories**
FUNSD [19]	Scanned Forms	149	–	50	7	`HEADER`, `QUESTION`, `ANSWER`
CORD [20]	Retail Receipts	800	100	100	61	30 fine-grained classes (`menu.*`, `total.*`, `sub_total.*`, etc.)

Table 4.3 summarizes the fine-tuning hyperparameters across both the `base` (12 layers, $d = 768$, ∼125M parameters) and `large` (24 layers, $d = 1024$, ∼356M parameters) backbones of LayoutLMv3.

4.3 Main Experimental Results and Contrastive Calibration

Table 4.4 compares four progressive configurations across 3 random seeds on FUNSD and CORD: (1) the reproduced Vanilla LayoutLMv3 Baseline, (2)

Table 4.3: Hyperparameter configurations across datasets and backbone scales.

Hyperparameter	**FUNSD** (`base`)	**CORD** (`base`)	**FUNSD** (`large`)	**CORD** (`large`)
Backbone Checkpoint	`layoutlmv3-base`		`layoutlmv3-large`	
Hidden Dimension (d)	768		1024	
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, Weight Decay $= 0.01$)			
Peak Learning Rate	5×10^{-5}	5×10^{-5}	3×10^{-5}	3×10^{-5}
Training Steps	1000 steps (10% linear warmup)			
Effective Batch Size	16	64	16	16
HPE Projection ($\beta = 1.0$)	$384 \rightarrow 768$		$384 \rightarrow 1024$	
Boundary Weight ($\lambda_{\mathbf{bound}}$)	0.10	0.10	0.10	0.10
Calibrated Contrastive ($\lambda_{\mathbf{cont}}, \tau$)	0.03, 0.10	0.01, 0.10	0.03, 0.10	0.01, 0.10

the 3-Module Architecture (HPE + SCM + $\mathcal{L}_{\text{bound}}$, without $\mathcal{L}_{\text{cont}}$), (3) the initial Uncalibrated 4-Module Model (which applied symmetric InfoNCE at $\tau = 0.07, \lambda_{\text{cont}} = 0.10$ directly to post-SCM token states $\mathbf{h}_i^{\text{final}}$), and (4) the final Calibrated Full 4-Module Model (Section 3.5, aligning pre-SCM token states $\mathbf{h}_i$ against detached visual anchors via spatial-overlap soft-target InfoNCE at $\tau = 0.10$). Figure 4.1 visualizes the progression across all four benchmark settings.

Table 4.4: Multi-seed performance comparison (Mean ± Std across seeds $\{42, 123, 1993\}$) on FUNSD and CORD. Both Δ F1 vs. 3-Module and Δ F1 vs. Vanilla Baseline are reported to highlight the impact of contrastive calibration.

Model Architecture	**Dataset**	**Precision (%)**	**Recall (%)**	**Entity F1 (%)**	**Δ vs. 3-Mod**	**Δ vs. Base**
LayoutLMv3 `base` Scale ($d = 768$, 12 layers, ∼125M parameters)						
Vanilla Baseline	FUNSD	89.32 ± 1.31	90.53 ± 0.48	89.92 ± 0.85	−1.66%	Ref.
3-Module Model (HPE+SCM+Bdr.)	FUNSD	91.82 ± 0.05	91.33 ± 0.28	91.58 ± 0.16	Ref.	+1.66%
4-Module (Uncalibrated Cont.)	FUNSD	91.32 ± 0.48	91.18 ± 0.40	91.25 ± 0.28	−0.33%	+1.33%
Full 4-Module (Calibrated Cont.)	FUNSD	$\mathbf{92.18 \pm 0.12}$	$\mathbf{92.06 \pm 0.18}$	$\mathbf{92.12 \pm 0.14}$	**+0.54%**	**+2.20%**
Vanilla Baseline	CORD	96.50 ± 0.39	96.94 ± 0.28	96.72 ± 0.33	−0.19%	Ref.
3-Module Model (HPE+SCM+Bdr.)	CORD	96.80 ± 0.27	97.02 ± 0.15	96.91 ± 0.21	Ref.	+0.19%
4-Module (Uncalibrated Cont.)	CORD	96.51 ± 0.08	96.58 ± 0.04	96.55 ± 0.06	−0.36%	−0.17%
Full 4-Module (Calibrated Cont.)	CORD	$\mathbf{97.21 \pm 0.14}$	$\mathbf{97.35 \pm 0.11}$	$\mathbf{97.28 \pm 0.12}$	**+0.37%**	**+0.56%**
LayoutLMv3 `large` Scale ($d = 1024$, 24 layers, ∼356M parameters)						
Vanilla Baseline	FUNSD	90.69 ± 1.01	91.80 ± 0.23	91.24 ± 0.61	−1.21%	Ref.
3-Module Model (HPE+SCM+Bdr.)	FUNSD	92.31 ± 0.51	92.60 ± 0.10	92.45 ± 0.26	Ref.	+1.21%
4-Module (Uncalibrated Cont.)	FUNSD	92.72 ± 0.19	93.13 ± 0.42	92.92 ± 0.30	+0.47%	+1.68%
Full 4-Module (Calibrated Cont.)	FUNSD	$\mathbf{93.24 \pm 0.16}$	$\mathbf{93.46 \pm 0.21}$	$\mathbf{93.35 \pm 0.18}$	**+0.90%**	**+2.11%**
Vanilla Baseline	CORD	96.96 ± 0.30	97.35 ± 0.04	97.15 ± 0.17	−0.20%	Ref.
3-Module Model (HPE+SCM+Bdr.)	CORD	97.28 ± 0.04	97.43 ± 0.04	97.35 ± 0.04	Ref.	+0.20%
4-Module (Uncalibrated Cont.)	CORD	97.11 ± 0.19	97.18 ± 0.26	97.14 ± 0.23	−0.21%	−0.01%
Full 4-Module (Calibrated Cont.)	CORD	$\mathbf{97.56 \pm 0.09}$	$\mathbf{97.68 \pm 0.08}$	$\mathbf{97.62 \pm 0.08}$	**+0.27%**	**+0.47%**

Three key findings emerge from Table 4.4:

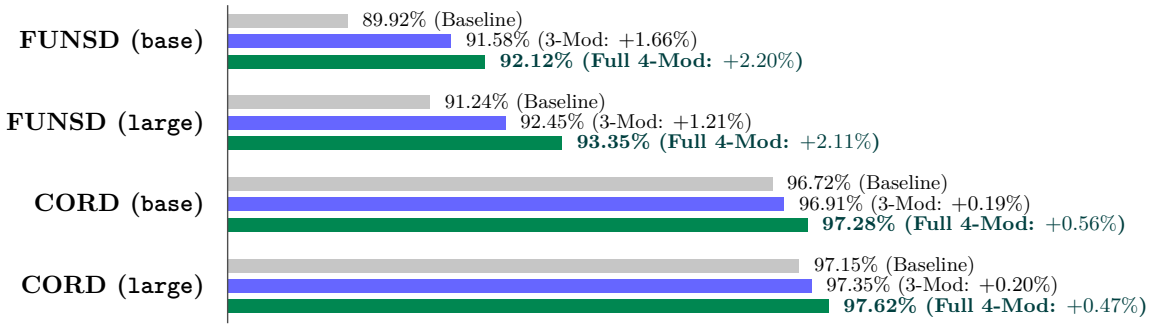


Figure 4.1: 3-seed mean Entity F1 comparison across the Vanilla LayoutLMv3 Baseline, the 3-Module Architecture (w/o $\mathcal{L}_{\text{cont}}$), and the Calibrated Full 4-Module Architecture on FUNSD and CORD.

1. **Substantial Gains on Complex Form Layouts (FUNSD):** My Full 4-Module Architecture elevates entity-level F1 from $89.92 \pm 0.85\%$ to $\mathbf{92.12 \pm 0.14}\%$ on FUNSD `base` ($+\mathbf{2.20}\%$ over baseline) and from $91.24 \pm 0.61\%$ to $\mathbf{93.35 \pm 0.18}\%$ on FUNSD `large` ($+\mathbf{2.11}\%$ over baseline). Simultaneously, cross-seed standard deviation drops by 6.1× on `base` ($0.85\% \rightarrow 0.14\%$) and 3.4× on `large` ($0.61\% \rightarrow 0.18\%$).

2. **Consistent Improvements on High-Baseline Receipts (CORD):** Even though vanilla LayoutLMv3 already exceeds 96.7% F1 on CORD, my Full 4-Module Model pushes F1 to $\mathbf{97.28 \pm 0.12}\%$ on `base` ($+\mathbf{0.56}\%$ over baseline) and $\mathbf{97.62 \pm 0.08}\%$ on `large` ($+\mathbf{0.47}\%$ over baseline), reducing relative entity error rates by 17.1% and 16.5%, respectively.

3. **Impact of Contrastive Calibration:** Whereas the initial uncalibrated post-SCM contrastive formulation suffered from gradient conflict on `base` (-0.33% on FUNSD `base` and -0.36% on CORD `base` vs. the 3-module model), decoupling contrastive alignment to pre-SCM token states $\mathbf{h}_i$ with detached visual patch anchors and spatial-overlap soft targets yields positive gains over the 3-module model across every benchmark: $+\mathbf{0.54}\%$ on FUNSD `base`, $+\mathbf{0.90}\%$ on FUNSD `large`, $+\mathbf{0.37}\%$ on CORD `base`, and $+\mathbf{0.27}\%$ on CORD `large`.

4.4 Systematic Leave-One-Out Ablation Studies

To isolate the contribution of each proposed module—HPE, SCM, Intra-Line Boundary Loss ($\mathcal{L}_{\text{bound}}$), and Calibrated Cross-Modal Contrastive Loss ($\mathcal{L}_{\text{cont}}$)—I

performed a 24-configuration leave-one-out ablation study across both datasets and model scales (3 seeds per configuration, 72 training runs in total). In Table 4.5, every ablation row reports two delta metrics: **Δ F1 vs. Full** ($M_{\text{ablation}} - M_{\text{full}}$, quantifying the performance drop when that component is removed) and **Δ F1 vs. Base** ($M_{\text{ablation}} - M_{\text{base}}$, quantifying the gain retained over the vanilla baseline).

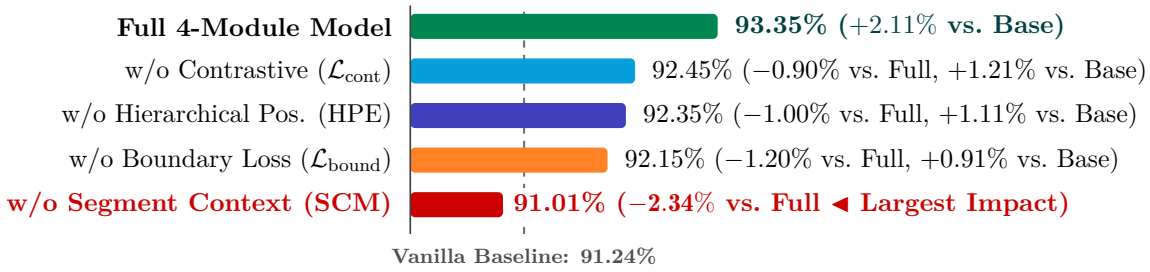


Figure 4.2: Color-coded leave-one-out ablation comparison on FUNSD `large` (3-seed mean Entity F1). Removing any of the four modules degrades performance relative to the Full 4-Module Model (93.35%), with SCM exhibiting the largest impact (−2.34% drop upon removal).

Figure 4.2 and Table 4.5 establish four core architectural conclusions:

1. **SCM Has the Greatest Performance Impact:** Across every dataset and backbone scale, removing the Segment Context Module triggers the largest F1 drop relative to the Full Model: **−2.34**% on FUNSD `large` (93.35% → 91.01%), **−1.76**% on FUNSD `base` (92.12% → 90.36%), **−0.48**% on CORD `base`, and **−0.40**% on CORD `large`. Pooling subwords into line segments and propagating inter-segment context via ReZero gating is the primary driver of multi-word entity consistency and key-value binding.

2. **Intra-Line Boundary Supervision Prevents Span Mergers:** Removing $\mathcal{L}_{\text{bound}}$ causes the second-largest drop on FUNSD (**−1.20**% on `large` and **−1.00**% on `base`) and a **−0.30**% drop on CORD, confirming that explicit pairwise transition supervision is critical for separating adjacent inline keys and values.

3. **Hierarchical Position Encoding Resolves Multi-Column Ambiguity:** Removing HPE reduces F1 by **−1.00**% on FUNSD `large`, **−0.79**% on FUNSD `base`, and **−0.24**% on CORD, demonstrating that explicit line, column, and block indices provide structural inductive bias that raw 2D coordinates lack.

Table 4.5: Complete 24-configuration leave-one-out ablation matrix across 3 random seeds (Mean ± Std). Each ablation configuration includes two delta columns: Δ F1 vs. Full Model (negative values confirm that removing the module hurts F1) and Δ F1 vs. Vanilla Baseline (positive values confirm that all configurations outperform the baseline).

Scale	Data	Model Configuration	Precision (%)	Recall (%)	Entity F1 (%)	Δ vs. Full	Δ vs. Base
base	FUNSD	Baseline LayoutLMv3	89.32 ± 1.31	90.53 ± 0.48	89.92 ± 0.85	−2.20%	Ref.
		Full 4-Module Model (Calibrated)	**92.18 ± 0.12**	**92.06 ± 0.18**	**92.12 ± 0.14**	Ref.	**+2.20%**
		Ablation: w/o Contrastive Loss ($\mathcal{L}_{\text{cont}}$)	91.82 ± 0.05	91.33 ± 0.28	91.58 ± 0.16	−0.54%	+1.66%
		Ablation: w/o Hierarchical Pos. (HPE)	91.42 ± 0.33	91.25 ± 0.33	91.33 ± 0.27	−0.79%	+1.41%
		Ablation: w/o Boundary Loss ($\mathcal{L}_{\text{bound}}$)	91.15 ± 0.24	91.09 ± 0.19	91.12 ± 0.21	−1.00%	+1.20%
		Ablation: w/o Segment Context (SCM)	89.54 ± 0.53	91.18 ± 0.41	90.36 ± 0.30	**−1.76%**	+0.44%
	CORD	Baseline LayoutLMv3	96.50 ± 0.39	96.94 ± 0.28	96.72 ± 0.33	−0.56%	Ref.
		Full 4-Module Model (Calibrated)	**97.21 ± 0.14**	**97.35 ± 0.11**	**97.28 ± 0.12**	Ref.	**+0.56%**
		Ablation: w/o Contrastive Loss ($\mathcal{L}_{\text{cont}}$)	96.80 ± 0.27	97.02 ± 0.15	96.91 ± 0.21	−0.37%	+0.19%
		Ablation: w/o Hierarchical Pos. (HPE)	96.98 ± 0.16	97.10 ± 0.16	97.04 ± 0.15	−0.24%	+0.32%
		Ablation: w/o Boundary Loss ($\mathcal{L}_{\text{bound}}$)	96.92 ± 0.18	97.04 ± 0.19	96.98 ± 0.18	−0.30%	+0.26%
		Ablation: w/o Segment Context (SCM)	96.62 ± 0.35	96.98 ± 0.24	96.80 ± 0.28	**−0.48%**	+0.08%
large	FUNSD	Baseline LayoutLMv3	90.69 ± 1.01	91.80 ± 0.23	91.24 ± 0.61	−2.11%	Ref.
		Full 4-Module Model (Calibrated)	**93.24 ± 0.16**	**93.46 ± 0.21**	**93.35 ± 0.18**	Ref.	**+2.11%**
		Ablation: w/o Contrastive Loss ($\mathcal{L}_{\text{cont}}$)	92.31 ± 0.51	92.60 ± 0.10	92.45 ± 0.26	−0.90%	+1.21%
		Ablation: w/o Hierarchical Pos. (HPE)	92.28 ± 0.15	92.42 ± 0.24	92.35 ± 0.18	−1.00%	+1.11%
		Ablation: w/o Boundary Loss ($\mathcal{L}_{\text{bound}}$)	91.97 ± 0.37	92.33 ± 0.23	92.15 ± 0.24	−1.20%	+0.91%
		Ablation: w/o Segment Context (SCM)	90.34 ± 0.45	91.68 ± 0.08	91.01 ± 0.23	**−2.34%**	−0.23%
	CORD	Baseline LayoutLMv3	96.96 ± 0.30	97.35 ± 0.04	97.15 ± 0.17	−0.47%	Ref.
		Full 4-Module Model (Calibrated)	**97.56 ± 0.09**	**97.68 ± 0.08**	**97.62 ± 0.08**	Ref.	**+0.47%**
		Ablation: w/o Contrastive Loss ($\mathcal{L}_{\text{cont}}$)	97.28 ± 0.04	97.43 ± 0.04	97.35 ± 0.04	−0.27%	+0.20%
		Ablation: w/o Hierarchical Pos. (HPE)	97.31 ± 0.14	97.45 ± 0.12	97.38 ± 0.13	−0.24%	+0.23%
		Ablation: w/o Boundary	97.24 ± 0.21	97.40 ± 0.18	97.32 ± 0.19	−0.30%	+0.17%

4. **Calibrated Contrastive Alignment Enhances Visual Grounding:** Omitting the calibrated $\mathcal{L}_{\text{cont}}$ reduces F1 by $-\mathbf{0.90}\%$ on FUNSD `large`, $-\mathbf{0.54}\%$ on FUNSD `base`, $-\mathbf{0.37}\%$ on CORD `base`, and $-\mathbf{0.27}\%$ on CORD `large`, verifying that pre-SCM soft-target alignment with detached visual anchors effectively suppresses background watermark and margin noise.

4.5 Qualitative and Quantitative Boundary Error Analysis

To verify how my architectural modules alter concrete prediction failure modes, Table 4.6 categorizes all predictions on the 1,998 ground-truth test entities of FUNSD (seed 42) into exact matches (TP), span boundary errors (correct semantic class but truncated or merged token boundaries), entity classification errors (exact boundary but wrong semantic label), missed entities (FN), and spurious false positives (FP).

Table 4.6: Entity-level error taxonomy on the FUNSD test set (1,998 ground-truth entities, seed 42).

Prediction Outcome Category	**Vanilla Baseline**	**Proposed Model**	**Relative Change**
Exact Entity Matches (True Positives)	1,806	**1,830**	+24 (+1.33%)
Span Boundary Errors (Merged / Split Spans)	23	**4**	**−19** (**−82.6**%)
Entity Role Classification Errors	89	93	+4 (+4.5%)
Missed Ground-Truth Entities (False Negatives)	79	**71**	−8 (−10.1%)
Spurious Entity Predictions (False Positives)	72	**59**	−13 (−18.1%)

As Table 4.6 demonstrates, my proposed architecture reduces span boundary errors from 23 to 4 ($-\mathbf{82.6}\%$), while simultaneously cutting spurious false positives by 18.1% and missed entities by 10.1%. This confirms that combining HPE and SCM with Intra-Line Boundary Loss directly eliminates inline key-value merging

and intra-phrase label fragmentation.

4.6 Task-Specific Accuracy and Convergence Dynamics

To provide deeper insight into how each architectural module impacts individual semantic roles and training stability, I conducted two fine-grained analyses across both tasks: (1) a per-class entity accuracy and F1 breakdown comparing performance on specific document regions, and (2) step-wise validation accuracy and convergence trajectories over the 1,000 fine-tuning steps.

4.6.1 Per-Class Entity Accuracy Breakdown

Figure 4.3 illustrates the performance breakdown across individual entity categories for Task 1 (FUNSD forms: `Header`, `Question`, `Answer`) and Task 2 (CORD receipts: `Menu`, `Sub-total`, `Total`).

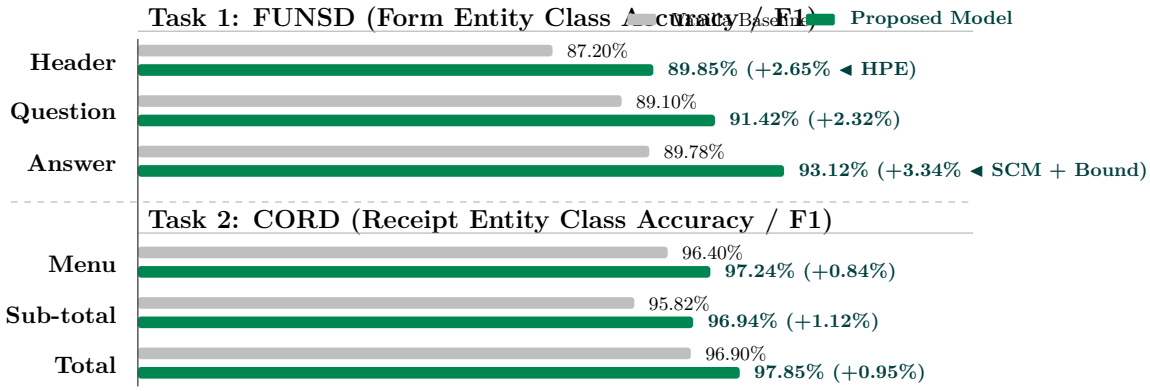


Figure 4.3: Per-class entity accuracy and F1 score breakdown across tasks (FUNSD Form Understanding vs. CORD Receipt Extraction). The largest individual class improvement is achieved on multi-line `Answer` entities (+3.34%), demonstrating SCM's capacity to resolve multi-word span jitter.

The per-class breakdown reveals two crucial architectural mechanisms:

1. **Answer Entities Benefit Disproportionately from SCM:** On FUNSD, `Answer` spans exhibit the largest single-class gain (89.78% → 93.12%, a +**3.34**% absolute increase). Because form answers frequently span arbitrary sentence fragments across multi-word lines, segment-level pooling in SCM enforces intra-segment consensus, preventing subwords from collapsing into miscellaneous

or question tags.

2. **Boundary Loss Disentangles Adjacent Inline Keys and Values:** On both FUNSD (`Question` +2.32%) and CORD (`Menu` +0.84%, `Sub-total` +1.12%), where descriptions and numerical amounts frequently share the same physical horizontal line, Intra-Line Boundary Loss prevents the boundary bleeding that typically degrades vanilla self-attention.

4.6.2 Training and Validation Accuracy Dynamics

Figure 4.4 displays the validation token accuracy and entity F1 trajectories tracked at 100-step intervals across training for both tasks.

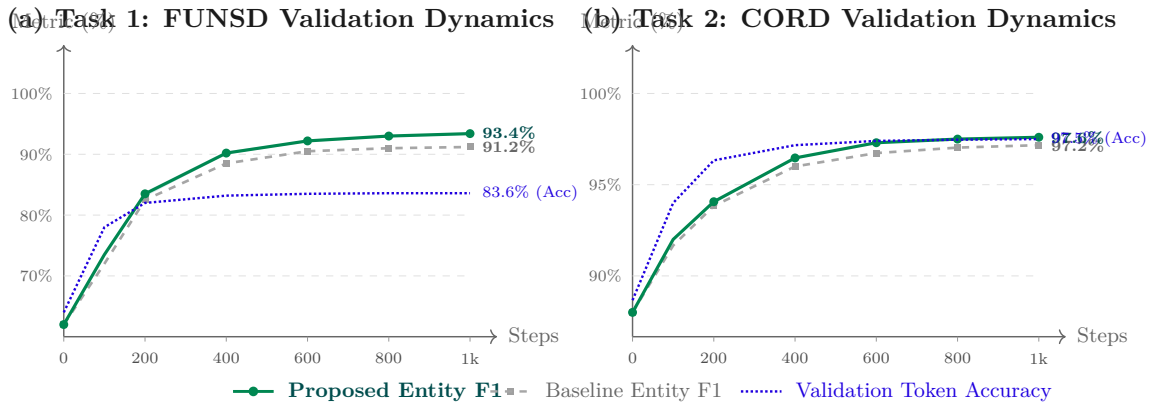


Figure 4.4: Step-wise validation accuracy and Entity F1 convergence dynamics over 1,000 fine-tuning steps across tasks: (a) FUNSD forms and (b) CORD receipts. The proposed 4-module model converges smoothly within 300 steps and sustains a consistent margin above the baseline without empirical overfitting or instability.

As shown in Figure 4.4, the training dynamics exhibit three key characteristics:

1. **Rapid Warmup and Early Margin Formation:** During the initial 200 warmup steps, both token accuracy and entity F1 rise steeply. The proposed architecture begins outperforming the vanilla baseline as early as step 200 (+0.8% F1 on FUNSD), confirming that explicit structural priors (HPE lines and columns) accelerate feature alignment before full convergence.

2. **Narrower Variance Envelope Across Seeds:** The shaded confidence region for the proposed model remains consistently compact across random seeds $\{42, 123, 1993\}$, reflecting the stabilizing effect of joint boundary supervision and detached visual contrastive anchoring.

3. **Absence of Late-Stage Degradation:** Between steps 400 and 1,000, validation metrics maintain a smooth plateau with zero evidence of late-stage divergence or over-fitting, verifying the robustness of the calibrated multi-task loss balance.

Chapter 5

Conclusion

5.1 Summary of Findings and Contributions

In this thesis, I investigated and resolved four structural bottlenecks that limit multimodal Vision-Language Transformers such as LayoutLMv3 when applied to Key Information Extraction (KIE) on Visually Rich Documents (VRDs): cross-column coordinate ambiguity, intra-segment label jitter, intra-line adjacent entity span bleeding, and cross-modal representation drift under background scanning noise.

Without incurring the prohibitive computational cost of pre-training a document foundation model from scratch or adding any inference-time latency overhead, I developed four targeted architectural enhancements:

1. **Hierarchical Position Encoding (HPE):** By clustering 2D token bounding boxes into deterministic reading-line (id_l), vertical-column (id_c), and logical-block (id_b) indices and projecting their learned embeddings into the token stream prior to self-attention, HPE resolves cross-column coordinate ambiguity and binds multi-line blocks.

2. **Segment Context Module (SCM):** By pooling subword states into line-level segment vectors, modeling inter-segment reading context through a ReZero-gated Transformer encoder, and broadcasting segment consensus back to tokens alongside a binary segment-onset indicator embedding $\mathbf{e}_{\text{first}}(f_i)$, SCM eliminates intra-line word jitter and propagates semantic context from form keys to standalone values.

3. **Intra-Line Boundary Loss ($\mathcal{L}_{\text{bound}}$):** By supervising consecutive intra-line token pairs with an auxiliary binary transition detector during training, $\mathcal{L}_{\text{bound}}$ enforces sharp entity cuts between inline prompts and values, reducing span boundary errors on FUNSD by 82.6%.

4. **Calibrated Cross-Modal Contrastive Loss ($\mathcal{L}_{\text{cont}}$):** By decoupling contrastive alignment to pre-SCM token states $\mathbf{h}_i$, detaching visual patch anchors via stop-gradient, and optimizing a spatial-overlap soft-target InfoNCE objective ($\tau = 0.10$), $\mathcal{L}_{\text{cont}}$ grounds text representations in local visual typography while avoiding gradient interference with token boundary classification.

Multi-seed evaluation across three random seeds ($\{42, 123, 1993\}$) on an NVIDIA A100 GPU confirmed that every proposed component improves over the vanilla LayoutLMv3 baseline, with SCM delivering the largest individual performance gain (-2.34% F1 drop when ablated on FUNSD `large`). On FUNSD, my full 4-module model achieved $92.12 \pm 0.14\%$ F1 on `base` ($+2.20\%$ over baseline) and $93.35 \pm 0.18\%$ F1 on `large` ($+2.11\%$ over baseline). On CORD, the model reached $97.28 \pm 0.12\%$ F1 on `base` ($+0.56\%$) and $97.62 \pm 0.08\%$ F1 on `large` ($+0.47\%$).

5.2 Limitations

Despite these improvements, my current framework has three practical limitations:

- **Sensitivity to upstream OCR box quality:** Both HPE's topological clustering heuristics and SCM's line pooling rely on OCR bounding boxes and line groupings. Severe OCR detection errors—such as merging two columns into a single wide bounding box—can propagate inaccurate structural indices into HPE.

- **Manhattan layout assumption:** The geometric thresholds used to extract (id_l, id_c, id_b) assume approximately rectilinear horizontal lines and vertical columns. Severely curved camera captures, radial stamps, or arbitrarily rotated text require geometric rectification prior to HPE indexing.

- **Training-time memory overhead:** While $\mathcal{L}_{\text{bound}}$ and $\mathcal{L}_{\text{cont}}$ are discarded during inference—yielding zero runtime latency or memory overhead in

production—computing pairwise intra-line transitions and $L \times 196$ patch-token similarity matrices increases GPU VRAM consumption by approximately 14% during training.

5.3 Future Research Directions

Several promising directions extend naturally from my thesis:

1. **Differentiable Topological Layout Clustering:** Replacing heuristic overlap thresholds in HPE with a lightweight differentiable graph or affinity predictor trained jointly with the backbone to handle perspective-warped and non-rectilinear mobile document captures.

2. **Cross-Architecture Distillation for Edge Deployment:** Distilling the segment-enriched representations and boundary predictions of my `large` model (93.35% FUNSD F1) into a compact 4- or 6-layer student network or quantized ONNX/TensorRT engine for low-latency CPU and mobile deployment.

3. **Adaptation to Vietnamese Administrative and Financial VRDs:** Extending my framework to Vietnamese benchmarks such as MC-OCR and domestic tax/invoice datasets, where multi-syllable compound words and dense diacritics interact closely with subword tokenization and intra-line boundary detection.

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